

Impacts of Artificial Intelligence

KEDGE - Data Analytics for Business

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Today

- Depending on the time available: go back over the texts and the questions
- Group work around AI use cases (our own AI fight club)
- Discussion time + additional elements from O'Neil's book
- Time to work on the final presentation

Session 2 - Algorithms and society: acceptability, bias, data

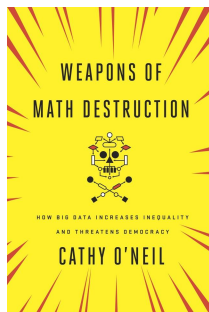
Reminder

Session 2 – Algorithms and society: acceptability, bias, data.

- O’NEIL C., 2016, « Weapons of Math Destruction », [Discover Magazine](#), 1 septembre 2016.
- THOMPSON S.A., WARZEL C., 2019, « Twelve Million Phones, One Dataset, Zero Privacy », [The New York Times](#), 19 décembre 2019.
- *Complementary reading:* O’NEIL C., 2016, Weapons of math destruction: how big data increases inequality and threatens democracy, New York, Crown.

O'NEIL C., 2016, Weapons of math destruction

How big data increases inequality and
threatens democracy



WMDs: Weapons of math
destruction

Dark side of big data

Selection of candidates

Justice

Perspectives (targeted citizen, etc.)

Cathy O'NEIL - Math, data science,



Mathematical background

- Phd from Harvard (1999)
- Postdoc at MIT
- Tenure-track professor at Barnard College (with Columbia University, NY)

Data scientist : from **university** to **industry**

- Finance and hedge fund D.E. Shaw(2007)
- 2008 financial crisis

Occupy wall street

Algorithmic Auditing

- ORCAA (2017?): O'Neil Risk Consulting & Algorithmic Auditing
- Bias and audit algorithms (Bias, gender, inequality, racial, etc.)

What are WMDs?

“Nevertheless, many of these models encoded human prejudice, misunderstanding, and bias into the software systems that increasingly managed our lives. Like gods, these mathematical models were **opaque**, their workings **invisible** to all but the highest priests in their domain: mathematicians and computer scientists. Their verdicts, even when **wrong** or **harmful**, were **beyond dispute or appeal**. And they tended to **punish** the poor and the oppressed in our society, while making the rich richer. I came up with a name for these harmful kinds of models: **Weapons of Math Destruction**, or **WMDs** for short.”

WMDs: Weapons of math destruction

WMDs 3 characteristics:

- **opacity:** opaque or invisible
 - proprietary / black-box
- **scale:** affect large number of people
 - “Capacity to grow exponentially” / “They can leap from one field to the next” (p.31)
- **damage:** negative effect
 - harms people, bias, racism/sexism, etc.

More about **algorithms** and big data than AI

- not necessarily AI
- but at least the issue of algorithms and (big) data
- “Dark side of big data”



Artificial
Intelligence



Seems to ring a bell ...

putt out of sight / liability / accountability (» Crawford)

- No room to complain/correct system error
- “Worse, losers in the unregulated e-score universe have little recourse to complain, much less correct the system’s error. In the realm of WMDs, they’re collateral damage.”

AI/algorithm = not neutral

- “Algorithms are opinions embedded in code”

Algorithms are opinions embedded in code

Liberation interview

You say that algorithms are opinions embedded in code ...

“Yes, because there is always a **definition of the conditions for success for the person who owns the algorithm**. And the question we have to ask ourselves is: does it also correspond to a success for me, who is targeted by this program? But **we have different perspectives, there is no objective definition of success**.

The university ranking system, for example, works very well for the people in charge of the universities. Their job is even to improve that score. So **the success defined by the model is linked to their own achievement**. But that's not the case for the students who go into debt, nor for the parents of students who pay for education. And it's not an achievement for society at large either. We want universities to be primarily a vehicle for upward mobility, for people to have a better life through education.

In the end, the only certainty is that it is a success for the person who designed the algorithm.”

In summary

“**Weapons of math destruction**, which O’Neil refers to throughout the book as WMDs, are **mathematical models or algorithms** that claim to quantify important traits: teacher quality, recidivism risk, creditworthiness but have harmful outcomes and often reinforce inequality, keeping the poor poor and the rich rich. They have **three things in common: opacity, scale, and damage**. They are often **proprietary** or otherwise **shielded from prying eyes**, so they have the effect of being a **black box**. They affect **large numbers of people**, increasing the chances that they get it wrong for some of them. And they have a **negative effect** on people, perhaps by encoding racism or other **biases** into an algorithm or enabling predatory companies to advertise selectively to vulnerable people, or even by causing a global financial crisis.”

Evelyn Lamb, [Review: Weapons of Math Destruction](#), Scientific American

EXERCISE

EXERCISE - AI fight club

One topic : justice & AI

Opposition: group for AI *versus* group against AI

Group work : I will give you a "position"/role (advocate vs. accuser)



EXERCISE

To get started: for inspiration and an overview of what can be done, read O'neil's article.

- O'NEIL C., 2016, « Weapons of Math Destruction », [Discover Magazine](#), 1 septembre 2016.
- See how she specifies the interests but also points out the risks and limitations

According to the assigned position (defender/critic) : **internet and documentary research** on the arguments for/against AI on the assigned topic

**Sharing of results,
confrontation, discussion and
exchange**

What does O'Neil say about that?

Additional elements / Further information

Justice and AI

- Chapter 5 - CIVILIAN CASUALTIES: Justice in the age of Big Data
- (I use the other examples from Chapter 1 and Afterword)

Justice & Algorithms

Duane Buck

Recidivism (and race)

LSI-R

Compas

Metrics and evaluation ?

Duane Buck case

Duane Buck case (1997)

- Killed 2 people
- Death penalty ?
- Fear of recidivism question
- African American
- Psychologist Walter Quijano
- Studied recidivism rates

Walter Quijano

Quijano, who had studied recidivism rates in the Texas prison system, made a reference to Buck's race, and during cross-examination the prosecutor jumped on it.

“You have determined that the...the race factor, black, increases the future dangerousness for various complicated reasons. Is that correct?” the prosecutor asked.

“Yes,” Quijano answered. The prosecutor stressed that testimony in her summation, and the jury sentenced Buck to death.

Race influence in sentencing

Challenge to several decisions in which WQ had testified and mentioned race

- “Six of the prisoners got new hearings but were again sentenced to death. Quijano’s prejudicial testimony, the court ruled, had not been decisive. Buck never got a new hearing, perhaps because it was his own witness who had brought up race. He is still on death row.”

Race and sentencing

- 20 percent longer sentences for black men than whites (similar crimes)
- 13 percent of the population, 40 percent of America’s prison cells

Using algorithm to neutralize sentencing bias (race, etc.) ?

Recidivism models

Tools helping judges to evaluate recidivism risk

Improvement:

- more consistent sentences
- less influence of the biases of judges
- save money (shorter average sentence)

“The **question**, however, is whether we’ve **eliminated human bias** or **simply camouflaged it with technology.**”

- Assumptions embedded in code
- “And while Walter Quijano’s words were transcribed for the record, which could later be read and challenged in court, the workings of a recidivism model are tucked away in algorithms, intelligible only to a tiny elite.”

LSI-R: Level of Service Inventory-Revised

The Level of Service Inventory–Revised™ (LSI-R™) is a **quantitative survey** of **offender attributes** and their situations relevant to level of supervision and treatment decisions.

Designed for ages 16 and older, the LSI-R helps **predict** parole outcome, success in correctional halfway houses, institutional misconducts, and recidivism.

The **54 items** included in this tool are based on legal requirements and include relevant factors needed for making decisions about risk and treatment.

LSI-R

Problem of “not so neutral” questions

- first time you were ever involved with the police ?
- friends and relatives with criminal records ?
- ➤ W/B , rich/poors

Feedback loop risk

Compas

Compas: Correctional Offender Management Profiling for Alternative Sanctions

- recidivism score
- predictions of “Risk of Recidivism”, “Risk of Violent Recidivism” etc.
- 127 criteria (?)

ProPublica audit (and others)

- not so good

Which metrics/indicators?

% of high-risk scores for black and whites matches with observed rates

- “common way of describing fairness”
- “equally likely to be right about blacks and whites considered as high-risk

BUT ... when it's wrong

- **2*false positives for blacks / 2*false negative for whites**
- “... blacks were twice as likely to be labeled high-risk but not return to prison as whites, and whites were twice as likely to be labeled as low risk but actually end up back in prison [...] as blacks”

ProPublica

“Our analysis of Northpointe’s tool, called COMPAS (which stands for Correctional Offender Management Profiling for Alternative Sanctions), found that black defendants were far more likely than white defendants to be incorrectly judged to be at a higher risk of recidivism, while white defendants were more likely than black defendants to be incorrectly flagged as low risk.”

<https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>

Measure and evaluation

What is a good metric ? In which case ?

- Example : avoid FP vs FN ?

Depending on the criteria used to evaluate the performance, the final picture is not at all the same.

+ risk of Goodhart's law: "When a measure becomes a target, it ceases to be a good measure"

» Same in medical AI and other fields

Group work on final presentation

Evaluation = Oral presentation

- On a theme of your choice related to the course (if justified, a case study can be done)
- In **groups of ~5** depending on the number of participants): ideally around 6-7 groups
- **~15 minutes** (depending on the number of groups)

Expectations / guidelines - 1

Gather documentation:

- At least **two** articles from the national/international **press**
 - French examples: Le Figaro, Le Monde, Libération, etc.
 - International examples: NYT, The Washington Post, El Pais, etc.
- At least **one** article from **academic** journals that will help you organize your reasoning
- At least **one book** (no obligation to read it in its entirety, but at least rely on it + you can consult reading notes in academic journals or social science sites)
- Also possible (no obligation): statistics (INSEE, Eurostat, etc.), radio programs / podcasts

Expectations / guidelines - 2

Construct a problematic

- Propose a presentation that is not a juxtaposition of descriptive elements but a rigorous and coherent argumentation allowing to answer the initial question.

Format: PowerPoint (or other presentation tools)

- Propose a presentation based on a rich and neat PowerPoint. Write down the essential arguments
- Insert a few significant quotations if necessary
- Bibliography : specify at the end of the PowerPoint the documentation used

Duration

- Each presentation lasts 15-20 minutes and is followed by a 10-minute discussion.